



TRIOS CYBER

LABORATORY PRACTICE 12

Authentication Event Search, Time Filtering, and Visualization with Splunk

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1. Splunk Installation on Ubuntu

Splunk Enterprise was installed on the Ubuntu virtual machine used for the laboratory. The Linux AMD64 package was downloaded from the Splunk download service and installed with the APT package manager. The package installation prepared the local system for the log-analysis activity.

```
matthew@matthew-VirtualBox:~$ wget -O splunk-10.4.3-4174a2deda5d-linux-amd64.deb "https://download.splunk.com/products/splunk/releases/10.4.3/linux/splunk-10.4.3-4174a2deda5d-linux-amd64.deb"
2026-09-13 18:28:21 -- https://download.splunk.com/products/splunk/releases/10.4.3/linux/splunk-10.4.3-4174a2deda5d-linux-amd64.deb
Resolving download.splunk.com (download.splunk.com)... 108.138.239.71, 108.138.239.54, 108.138.239.105, ...
Conectando con download.splunk.com (download.splunk.com)[108.138.239.71]:443... conectado.
Petición HTTP enviada, esperando respuesta... 200 OK
Longitud: 1324052468 (1,2G) [binary/octet-stream]
Guardando como: "splunk-10.4.3-4174a2deda5d-linux-amd64.deb"

splunk-10.4.3-4174a2deda5d 100%[=====] 1,23G 59,6MB/s en 29s

2026-09-13 18:28:50 (43,7 MB/s) - "splunk-10.4.3-4174a2deda5d-linux-amd64.deb" guardado [1324052468/1324052468]

matthew@matthew-VirtualBox:~$ ls -lh splunk*.deb
-rw-r--r-- 1 matthew matthew 1,3G ago 31 11:50 splunk-10.4.3-4174a2deda5d-linux-amd64.deb
```

```
matthew@matthew-VirtualBox:~$ sudo apt install ./splunk-10.4.3-4174a2deda5d-linux-amd64.deb -y
Leyendo lista de paquetes... Hecho
Creando árbol de dependencias
Leyendo la información de estado... Hecho
Nota, seleccionando «splunk» en lugar de «./splunk-10.4.3-4174a2deda5d-linux-amd64.deb»
El paquete indicado a continuación se instaló de forma automática y ya no es necesario.
libwacom-bin
Utilice «sudo apt autoremove» para eliminarlo.
Se instalarán los siguientes paquetes NUEVOS:
splunk
0 actualizados, 1 nuevos se instalarán, 0 para eliminar y 564 no actualizados.
Se necesita descargar 0 B/1,324 MB de archivos.
Se utilizarán 0 B de espacio de disco adicional después de esta operación.
Des:1 /home/matthew/splunk-10.4.3-4174a2deda5d-linux-amd64.deb splunk amd64 10.4.3 [1,324 MB]
Seleccionando el paquete splunk previamente no seleccionado.
(Leyendo la base de datos ... 187462 ficheros o directorios instalados actualmente.)
Preparando para desempaquetar .../splunk-10.4.3-4174a2deda5d-linux-amd64.deb ...
Verify that this system has all the commands we will require to perform the preflight step
no need to run the splunk-preinstall upgrade check
Desempaquetando splunk (10.4.3) ...
Configurando splunk (10.4.3) ...
find: '/opt/splunk/lib/python3.7/site-packages': No existe el archivo o el directorio
complete
```

2. Splunk Initialization and Web Interface Access

Splunk was started from /opt/splunk/bin. During the first launch, administrator credentials were created and the platform completed its initial checks before starting the splunkd service. Splunk Web became available on port 8000 and was opened in Firefox, confirming that the analysis platform was operational.

```
matthew@matthew-VirtualBox:~$ sudo /opt/splunk/bin/splunk start --accept-license --run-as-root
This appears to be your first time running this version of Splunk.

Splunk software must create an administrator account during startup. Otherwise, you cannot log in.
Create credentials for the administrator account.
Characters do not appear on the screen when you type in credentials.

Please enter an administrator username: matthew
Password must contain at least:
  * 8 total printable ASCII character(s).
Please enter a new password:
Please confirm new password:
ERROR: Password did not meet complexity requirements. Password must contain at least:
  * 8 total printable ASCII character(s).
Please enter a new password:
Please confirm new password:
ERROR: Password did not meet complexity requirements. Password must contain at least:
  * 8 total printable ASCII character(s).
Please enter a new password:
Please confirm new password:
Copying '/opt/splunk/etc/openldap/ldap.conf.default' to '/opt/splunk/etc/openldap/ldap.conf'.
writing RSA key
writing RSA key

Moving '/opt/splunk/share/splunk/search_mrsparkle/modules.new' to '/opt/splunk/share/splunk/search_mrsparkle/modules'.

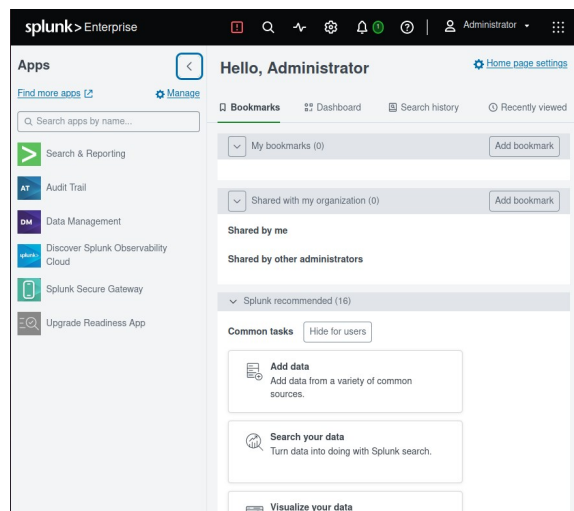
Splunk> Another one.

Checking prerequisites...
Checking http port [8000]: open
Checking mport port [8089]: open
Checking appserver port [127.0.0.1:8065]: open
Checking kvstore port [8191]: open
Checking configuration... Done.
Creating: /opt/splunk/var/lib/splunk
Creating: /opt/splunk/var/run/splunk/appserver/118n
Creating: /opt/splunk/var/run/splunk/appserver/modules/static/css
Creating: /opt/splunk/var/run/splunk/upload
Creating: /opt/splunk/var/run/splunk/search_telemetry
Creating: /opt/splunk/var/run/splunk/search_log
Creating: /opt/splunk/var/spool/splunk
```

```
Creating: /opt/splunk/var/run/splunk/search_log
Creating: /opt/splunk/var/spool/splunk
Creating: /opt/splunk/var/spool/dirmoncache
Creating: /opt/splunk/var/lib/splunk/authob
Creating: /opt/splunk/var/lib/splunk/hashob
Creating: /opt/splunk/var/run/splunk/collect
Creating: /opt/splunk/var/run/splunk/sessions
New certs have been generated in '/opt/splunk/etc/auth'.
New certs have been generated in '/opt/splunk/etc/auth'.
Checking critical directories... Done
Checking indexes...
Validated: audit_configtracker_dm_summary_dsapvennt_dsclient_dsphonehome
_internal_introspection_metrics_metrics_rollup_telemetry_thefishbucket history main summary
Done
Checking filesystem compatibility... Done
Checking conf files for problems... Done
Checking default conf files for edits...
Validating installed files against hashes from '/opt/splunk/splunk-10.4.3-4174a2deda5d-linux-amd64-manifest'
All installed files intact.
Done
All preliminary checks passed.

Starting splunk server daemon (splunkd)...
Using configuration from /opt/splunk/share/openssl3/openssl.cnf
.....
Certificate request self-signature ok
subject=C=matthew-VirtualBox, O=splunkUser
Done

Waiting for web server at http://127.0.0.1:8000 to be available.....
```



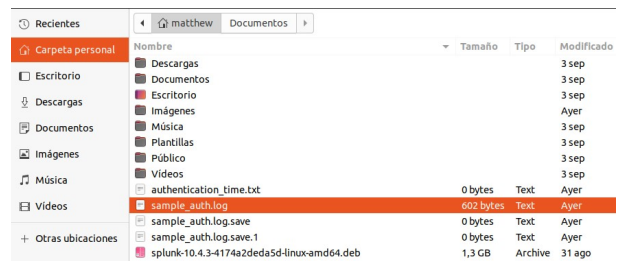
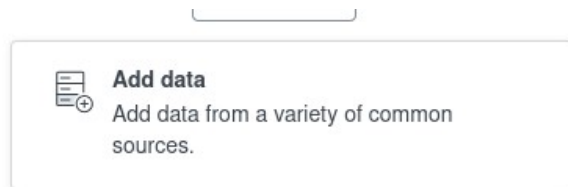
3. Authentication Sample Data Preparation

A small authentication-style log file named `sample_auth.log` was prepared on Ubuntu and displayed with `cat` before ingestion. The file contained seven timestamped records with the fields `user`, `src`, `result`, and `message`. The dataset included both FAILED and SUCCESS authentication outcomes for the users `matthew`, `admin`, and `test`.

```
matthew@matthew-VirtualBox:~$ cat sample_auth.log
2026-09-12T13:00:01 user=matthew src=10.10.10.20 result=FAILED message=Failed_password
2026-09-12T13:00:05 user=matthew src=10.10.10.20 result=FAILED message=Failed_password
2026-09-12T13:00:10 user=admin src=10.10.10.30 result=FAILED message=Failed_password
2026-09-12T13:00:15 user=matthew src=10.10.10.20 result=SUCCESS message=Accepted_password
2026-09-12T13:00:20 user=admin src=10.10.10.30 result=FAILED message=Failed_password
2026-09-12T13:00:25 user=test src=10.10.10.40 result=FAILED message=Invalid_user
2026-09-12T13:00:30 user=matthew src=10.10.10.20 result=FAILED message=Failed_password
```

4. Data Upload and Source Type Configuration

The sample log was added to Splunk through the Add Data workflow. The file was selected from the Ubuntu user directory and previewed before indexing. A custom source type named `sample_auth` was used so the uploaded authentication records could be identified consistently during the search activity.



splunk-enterprise Apps

Administrator Messages Settings Activity Help Find

Add Data Select Source Set Source Type Input Settings Review Done < Back Next >

Set Source Type

This page lets you see how the Splunk platform sees your data before indexing. If the events look correct and have the right timestamps, click "Next" to proceed. If not, use the options below to define proper event breaks and timestamps. If you cannot find an appropriate source type for your data, create a new one by clicking "Save As".

Source: `sample_auth.log` [View Event Summary](#)

Source type: Select Source Type Save As

Time	Event
9/12/26 1:00:01.000 PM	2026-09-12T13:00:01 user=matthew src=10.10.10.20 result=FAILED message=Failed_password
9/12/26 1:00:05.000 PM	2026-09-12T13:00:05 user=matthew src=10.10.10.20 result=FAILED message=Failed_password
9/12/26 1:00:10.000 PM	2026-09-12T13:00:10 user=admin src=10.10.10.30 result=FAILED message=Failed_password
9/12/26 1:00:15.000 PM	2026-09-12T13:00:15 user=matthew src=10.10.10.20 result=SUCCESS message=Accepted_password
9/12/26 1:00:20.000 PM	2026-09-12T13:00:20 user=admin src=10.10.10.30 result=FAILED message=Failed_password
9/12/26 1:00:25.000 PM	2026-09-12T13:00:25 user=test src=10.10.10.40 result=FAILED message=Invalid_user
9/12/26 1:00:30.000 PM	2026-09-12T13:00:30 user=matthew src=10.10.10.20 result=FAILED message=Failed_password

Format Select... Select...

Event Breaks
Timestamp
Advanced

CTRL DERECHA

5. Input Settings and Data Review

The input configuration assigned the host value matthew-VirtualBox and used the default Splunk index configuration. The Review screen confirmed the uploaded file name sample_auth.log, the sample_auth source type, the host value, and the selected index before the data was submitted.

Input Settings
Optionally set additional input parameters for this data input as follows:

Host
When the Splunk platform indexes data, each event receives a "host" value. The host value should be the name of the machine from which the event originates. The type of input you choose determines the available configuration options. [Learn More](#)

Host field value:

☒ Constant value
☐ Regular expression on path
☐ Segment in path

Index
The Splunk platform stores incoming data as events in the selected index. Consider using a "sandbox" index as a destination if you have problems determining a source type for your data. A sandbox index lets you troubleshoot your configuration without impacting production indexes. You can always change this setting later. [Learn More](#)

FAQ
 > How do indexes work?
 > How do I know when to create or use multiple indexes?

Review

Input Type: Uploaded File
 File Name: sample_auth.log
 Source Type: sample_auth
 Host: matthew-VirtualBox
 Index: Default

6. Authentication Event Search

Search & Reporting was opened and the indexed data was queried for authentication outcomes. The search selected records containing result=FAILED or result=SUCCESS, allowing both unsuccessful and successful events from the sample file to be reviewed in one result set.

New Search
index=main | result=FAILED OR result=SUCCESS

7 events (before 9/13/26 7:01:33.000 PM) No Event Sampling

Time	Event
9/12/26 1:00:30.000 PM	2026-09-12T13:00:30 user=matthew src=10.10.10.20 result=FAILED message=Failed_password host = matthew-VirtualBox source = sample_auth.log sourcetype = sample_auth
9/12/26 1:00:25.000 PM	2026-09-12T13:00:25 user=test src=10.10.10.40 result=FAILED message=Invalid_user host = matthew-VirtualBox source = sample_auth.log sourcetype = sample_auth
9/12/26 1:00:20.000 PM	2026-09-12T13:00:20 user=admin src=10.10.10.30 result=FAILED message=Failed_password host = matthew-VirtualBox source = sample_auth.log sourcetype = sample_auth
9/12/26 1:00:15.000 PM	2026-09-12T13:00:15 user=matthew src=10.10.10.20 result=SUCCESS message=Accepted_password host = matthew-VirtualBox source = sample_auth.log sourcetype = sample_auth
9/12/26 1:00:10.000 PM	2026-09-12T13:00:10 user=admin src=10.10.10.30 result=FAILED message=Failed_password host = matthew-VirtualBox source = sample_auth.log sourcetype = sample_auth
9/12/26 1:00:05.000 PM	2026-09-12T13:00:05 user=matthew src=10.10.10.20 result=FAILED message=Failed_password host = matthew-VirtualBox source = sample_auth.log sourcetype = sample_auth
9/12/26 1:00:01.000 PM	2026-09-12T13:00:01 user=matthew src=10.10.10.20 result=FAILED message=Failed_password host = matthew-VirtualBox source = sample_auth.log sourcetype = sample_auth

7. Time Filtering

A custom Date & Time Range was applied to constrain the search to the period containing the authentication records. The filtered event list remained focused on the sample activity while demonstrating how Splunk can limit investigations to a defined time window.

Presets
 > Relative
 > Real-time
 > Date Range

Date & Time Range
 Between 09/12/2026 1:00:10.000 and 09/13/2026 1:00:30.000
 HH:MM:SS.SSS HH:MM:SS.SSS

Apply

New Search
index=main | result=FAILED OR result=SUCCESS

7 events (before 9/13/26 7:01:33.000 PM) No Event Sampling

Time	Event
9/12/26 1:00:30.000 PM	2026-09-12T13:00:30 user=matthew src=10.10.10.20 result=FAILED message=Failed_password host = matthew-VirtualBox source = sample_auth.log sourcetype = sample_auth
9/12/26 1:00:25.000 PM	2026-09-12T13:00:25 user=test src=10.10.10.40 result=FAILED message=Invalid_user host = matthew-VirtualBox source = sample_auth.log sourcetype = sample_auth
9/12/26 1:00:20.000 PM	2026-09-12T13:00:20 user=admin src=10.10.10.30 result=FAILED message=Failed_password host = matthew-VirtualBox source = sample_auth.log sourcetype = sample_auth
9/12/26 1:00:15.000 PM	2026-09-12T13:00:15 user=matthew src=10.10.10.20 result=SUCCESS message=Accepted_password host = matthew-VirtualBox source = sample_auth.log sourcetype = sample_auth
9/12/26 1:00:10.000 PM	2026-09-12T13:00:10 user=admin src=10.10.10.30 result=FAILED message=Failed_password host = matthew-VirtualBox source = sample_auth.log sourcetype = sample_auth
9/12/26 1:00:05.000 PM	2026-09-12T13:00:05 user=matthew src=10.10.10.20 result=FAILED message=Failed_password host = matthew-VirtualBox source = sample_auth.log sourcetype = sample_auth
9/12/26 1:00:01.000 PM	2026-09-12T13:00:01 user=matthew src=10.10.10.20 result=FAILED message=Failed_password host = matthew-VirtualBox source = sample_auth.log sourcetype = sample_auth

8. Failed Authentication Statistics by User

A second search extracted the user and result fields with rex, retained only result=FAILED records, and summarized them with stats count by user. The results showed three failed events for matthew, two for admin, and one for test, for a total of six failed authentication records.

New Search Save As Create Table View Close

Time range: Date time range Q

```
index=main
| rex "user=(?<user>\S+)"
| rex "result=(?<result>\S+)"
| search result=FAILED
| stats count by user
```

✓ 6 events (9/12/26 1:00:10.000 AM to 9/13/26 1:00:30.000 AM) No Event Sampling

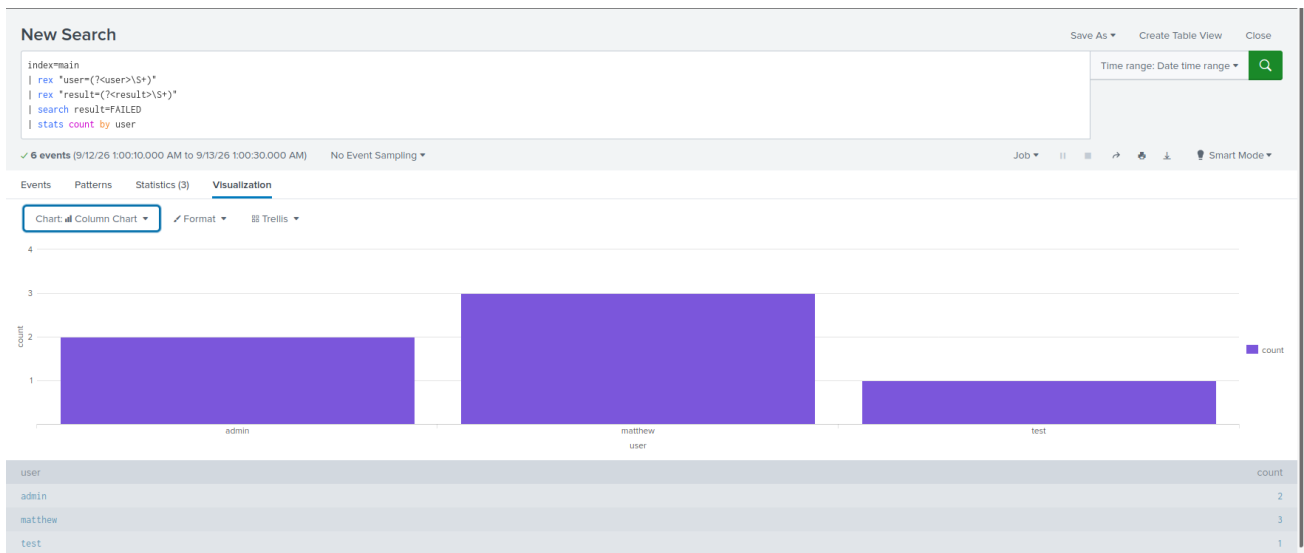
Events Patterns **Statistics (3)** Visualization

Show: 20 Per Page Format Preview: On

user	count
admin	2
matthew	3
test	1

9. Visualization Creation

The statistical results were converted to a column chart in the Visualization view. The chart provided a simple comparison of failed authentication counts by username and made matthew the highest-frequency account in the sample dataset visible at a glance.



10. Saved Search / Report

The completed analysis was saved as a report named Failed Authentication Events by User with the description "Count of failed authentication events grouped by username." The saved report preserved the search results and visualization for later review.

Save As Report ×

⚠ Your report has an absolute time range.

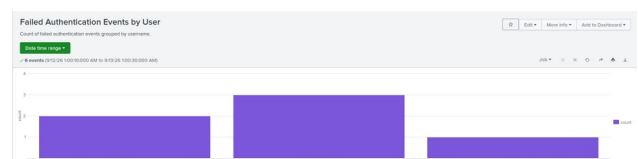
Title: Failed Authentication Events by User

Description: Count of failed authentication events grouped by username.

Content: Table Chart Table

Time Range Picker: Yes No

Cancel Save



11. Authentication Analysis Summary

The Splunk activity produced a compact authentication analysis workflow that combined ingestion, searching, time restriction, aggregation, visualization, and report saving. The main observed results are summarized below.

Field	Observed Value	Interpretation
Platform	Splunk Enterprise	Local log-analysis platform running on Ubuntu
Host	matthew-VirtualBox	System assigned to the uploaded events
Data source	sample_auth.log	Seven authentication-style records
Source type	sample_auth	Custom type assigned during ingestion
Authentication results	6 FAILED / 1 SUCCESS	Both unsuccessful and successful activity was searchable
Failed users	matthew: 3; admin: 2; test: 1	Frequency distribution produced with stats count by user
Time filtering	Custom Date & Time Range	Search results constrained to the relevant event window
Visualization	Column chart	Failed authentication events compared by username
Saved report	Failed Authentication Events by User	Reusable saved analysis and visualization

12. Laboratory Workflow Review

The laboratory followed a complete basic SIEM workflow: authentication records were prepared on the Ubuntu endpoint, uploaded into Splunk, assigned a source type and host, searched by authentication result, restricted by time, aggregated by username, visualized, and saved as a report. This sequence demonstrates how a log-analysis platform can turn raw authentication records into a concise security view suitable for initial investigation.

13. Conclusions

The Splunk log-analysis activity was successfully completed on the Ubuntu virtual machine. Authentication-related sample events were ingested and searched, a custom time filter was applied, and failed events were summarized by user. Six failed records and one successful record were available in the dataset, with matthew producing the highest failed-event count. A column chart was created and the analysis was saved as the report Failed Authentication Events by User. The exercise demonstrated the basic use of Splunk for authentication-event searching, time-based filtering, visualization, and reusable report creation.